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Providing Insights for Crisis Recovery in an Online Food Delivery Startup

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A dissertation submitted in partial fulfillment of the degree of Bachelor of Science

CERTIFICATE

This is to certify that **Mohd Ziyan, Sajid Khan, Mohd Aasif and Hamdan** has successfully completed the project entitled

"Providing Insights for Crisis Recovery in an Online Food Delivery Startup"

under the supervision **Prof. S. M. K. Quadri** in partial fulfillment of the requirements for the degree of **Bachelor of Science** at the **Department of Computer Science, Jamia Millia Islamia, New Delhi** during the academic years 2025–2026.

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DECLARATION

We hereby declare that the thesis entitled "**Providing Insights for Crisis Recovery in an Online Food Delivery Startup**" submitted in partial fulfilment of the requirements for the degree is my original work carried out under the guidance of our supervisor.

We further declare that this work has not been submitted, either in part or in full, to any other university or institution for the award of any degree, diploma, or certificate.

The data used in this research has been sourced from a structured dataset containing transaction-level records of a food delivery startup operating across eight major cities in India. All analyses, interpretations, and conclusions presented in this thesis are our own.

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ABSTRACT

The rapid proliferation of digital food delivery platforms in India has created an unprecedented volume of transactional, behavioural, and operational data. This thesis undertakes a rigorous and systematic data analytics investigation into the operations of a food delivery startup that serves eight major Indian cities — Bengaluru, Mumbai, Delhi, Chennai, Hyderabad, Pune, Ahmedabad, and Kolkata. The primary objective is to extract actionable business intelligence from a multi-dimensional dataset comprising 149,166 orders, 107,776 customers, 15,000 delivery partners, 19,995 restaurants, 342,671 menu items, and 68,842 customer ratings, spanning the period from January 2025 to September 2025.

The research employs a structured quantitative methodology, involving data extraction, cleaning, feature engineering, exploratory data analysis (EDA), and multi-dimensional business question analysis. Twelve core business questions are investigated, each mapped to a strategic operational domain: city-wise revenue performance, food category preferences, temporal order patterns, revenue trend analysis, cancellation behaviour, delivery performance, customer acquisition channel effectiveness, cuisine-based rating analytics, statistical correlations, vegetarian versus non-vegetarian preference segmentation, discount elasticity, and restaurant performance ranking.

Key findings reveal that Bengaluru generates the highest order volume (38,862 completed orders) and revenue (INR 1.37 crore), while Ahmedabad has the lowest cancellation rate (3.91%). The overall cancellation rate stands at 7.45%, representing 11,112 lost orders. The most critical operational challenge identified is the late delivery rate of 63.84%, meaning nearly two-thirds of all orders are delivered beyond the expected time window, with an average delay of 5.7 minutes over an average distance of 4.49 km. Beverages and Starters are the highest-selling categories. Night-time and Lunch slots (12:00–16:00 and 19:00–23:00) account for the majority of orders. The Organic customer acquisition channel contributes 50.8% of all orders. The average customer rating is a healthy 4.05 out of 5.

The thesis concludes with 15 detailed strategic recommendations across operational, marketing, technology, and restaurant partnership dimensions. These recommendations aim to reduce delivery delays, improve cancellation management, optimise marketing spend, and strengthen high-performing city and restaurant partnerships.

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The food delivery industry represents one of the most dynamic and data-rich segments of the modern digital economy. In India, the sector has witnessed explosive growth driven by smartphone penetration, increasing urbanisation, rising disposable incomes, and a cultural shift towards convenience-driven consumption. Platforms such as Zomato and Swiggy have collectively transformed how hundreds of millions of urban Indians access food, creating a marketplace that is simultaneously a logistics company, a technology platform, and a customer experience engine.

This transformation has generated an enormous volume of operational and transactional data — every order placed, every delivery completed or delayed, every rating submitted, and every discount applied creates a data point that, when analysed collectively, can reveal powerful insights about business performance, customer preferences, and operational efficiency. The capacity to turn this raw data into actionable intelligence constitutes the central value proposition of business analytics in the food-tech sector.

India's online food delivery market was valued at approximately USD 7.5 billion in 2023 and is projected to reach USD 21 billion by 2028, growing at a compound annual growth rate (CAGR) of approximately 23%. This growth is not uniform — it is concentrated in Tier-1 cities such as Bengaluru, Mumbai, and Delhi, which together account for a disproportionate share of order volume and revenue. However, Tier-2 cities are increasingly joining the digital consumption wave, creating new expansion opportunities for platform operators.

The competitive landscape of food delivery is intensely dynamic. Platforms compete on speed, variety, pricing, discounts, and user experience. A single negative experience — a delayed delivery, a cold meal, or an incorrect order — can reduce customer lifetime value significantly. This makes the ability to monitor, analyse, and continuously improve operational metrics not merely an academic exercise but a core competitive capability.

This thesis focuses on a food delivery startup that operates across eight major Indian cities: Bengaluru, Mumbai, Delhi, Chennai, Hyderabad, Pune, Ahmedabad, and Kolkata. The startup's operational dataset — spanning nine months from January to September 2025 —

provides a rich, multi-layered view of the business. By applying systematic data analytics techniques using Python, Pandas, NumPy, Matplotlib, and Seaborn, this research aims to transform this raw operational data into a comprehensive business intelligence report that can inform strategic decision-making.

We conducted this study in the context of QuickBite Express, a Bengaluru-based food delivery startup that experienced a significant platform crisis in mid-2025, making the need for rigorous data-driven analysis both timely and critical.

1.2 Statement of the Problem

QuickBite Express, a Bengaluru-based food-tech startup founded in 2020, faced a severe operational and reputational crisis in June 2025. A food safety incident at partner restaurants, amplified through social media, combined with a week-long delivery outage, triggered sharp customer disengagement and a significant decline in daily orders. Despite having access to large volumes of transactional data, the organisation lacked the structured analytical framework needed to diagnose these failures, guide recovery efforts, and prevent recurrence. This thesis addresses that gap.

Despite the availability of rich transactional data, QuickBite Express — like many food delivery startups in growth phases — struggled to extract structured, actionable insights. Decision-making remained reactive rather than proactive. This created several critical gaps:

- City-level performance is poorly understood: Revenue and order distribution across cities is not systematically monitored, leading to sub-optimal marketing resource allocation.
- Delivery failures are unquantified: The precise scale of late deliveries, the distribution of delays, and the relationship between distance and delivery time are rarely measured rigorously.
- Customer acquisition costs are poorly evaluated: Multiple acquisition channels (Organic, Paid, Referral, Social) are used simultaneously without rigorous ROI analysis.
- Menu and category performance is intuitive: Decisions about menu expansion, promotional focus, and food category prioritisation are made without data-driven category performance benchmarks.
- Discount strategy lacks analytical underpinning: Discounts are offered without systematic analysis of their impact on order frequency, basket size, and revenue contribution.

These gaps represent significant missed opportunities for operational improvement and revenue optimisation. This thesis addresses each of these gaps systematically through twelve business question analyses.

1.3 Objectives of the Study

The primary objectives of this thesis are:

1. To perform a comprehensive exploratory data analysis (EDA) of the food delivery startup's operational dataset.
2. To identify the city-wise distribution of orders and revenue and determine the top-performing markets.
3. To analyse food category and menu item performance to inform menu strategy.
4. To identify peak ordering periods by hour of day, day of week, and month.
5. To examine monthly revenue trends and identify growth patterns or seasonal variations.
6. To quantify the scale of order cancellations and identify problem cities and time slots.
7. To measure delivery performance, including the late delivery rate, average delay, and the relationship between distance and delivery time.
8. To evaluate the effectiveness of different customer acquisition channels in terms of order volume and revenue contribution.
9. To assess customer satisfaction through rating and sentiment score analysis across cuisine types.
10. To examine statistical correlations between key numerical variables in the dataset.
11. To compare vegetarian and non-vegetarian food preferences in terms of quantity sold, revenue, and average price.
12. To analyse the impact of discounts on order frequency and average order value.
13. To identify top-performing restaurant partners by revenue and order volume.
14. To formulate strategic, data-driven recommendations for improving business performance.

1.4 Scope and Limitations

This study is scoped to the operational dataset of a single food delivery startup operating in eight Indian cities over a nine-month period (January – September 2025). The analysis is primarily quantitative and descriptive in nature, with statistical correlations examined but causal relationships not formally established through econometric modelling.

The following limitations apply:

- The dataset represents approximately nine months of operations; seasonal effects beyond this window cannot be assessed.

- Customer demographic data beyond city of residence and acquisition channel is not available, limiting customer segmentation depth.
- The reasons for order cancellations are not captured in the dataset, precluding root cause analysis of cancellation behaviour.
- Delivery partner performance at the individual level is available but not linked to individual order outcomes in the delivered dataset, limiting partner-level attribution analysis.
- External factors such as weather, traffic, and city-specific events that may influence delivery performance are not captured.

1.5 Significance of the Study

This thesis makes several significant contributions to the academic and practical understanding of food delivery analytics in the Indian context:

- It provides a complete, replicable analytical framework — from raw data ingestion to business intelligence — that can serve as a template for similar studies.
- It generates empirical benchmarks for key performance indicators (KPIs) — such as a 63.84% late delivery rate and 7.45% cancellation rate — that are rare in publicly available research for Indian food-tech startups.
- It demonstrates the application of Python-based data analytics tools to a real-world business context, bridging the gap between academic methodology and industry practice.
- Its recommendations are directly actionable, addressing specific operational pain points identified through data analysis.

1.6 Organisation of the Thesis

This thesis is organised into seven chapters. Chapter 1 provides the introduction, background, objectives, and scope. Chapter 2 reviews relevant academic and industry literature on food delivery platforms, business intelligence, and customer analytics. Chapter 3 describes the research methodology and analytical framework. Chapter 4 presents the dataset in detail and documents the data preprocessing steps. Chapter 5 presents the exploratory data analysis. Chapter 6 contains the twelve business question analyses with findings and visualisation descriptions. Chapter 7 presents the consolidated findings, strategic recommendations, and conclusion. References and appendices are provided at the end.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Global Food Delivery Landscape

The global online food delivery industry has undergone a structural transformation over the past decade. What began as simple restaurant-to-door telephone ordering has evolved into a sophisticated digital marketplace ecosystem involving multi-sided platforms, real-time logistics networks, and algorithm-driven personalisation engines. Academic and industry research has tracked this evolution across multiple dimensions.

Parker, Van Alstyne, and Choudary (2016) in their seminal work 'Platform Revolution' established the theoretical framework for understanding multi-sided platform businesses like food delivery apps. They argue that such platforms create value by facilitating interactions between distinct user groups — restaurants, customers, and delivery partners — and that the key to competitive advantage lies in network effects. As the number of participating restaurants and delivery partners grows, the platform becomes more attractive to customers, creating a self-reinforcing cycle of growth.

Rochet and Tirole (2003) pioneered the economic theory of two-sided markets, which directly applies to food delivery platforms. Their analysis demonstrated that pricing on one side of the market (e.g., delivery fees charged to customers) affects participation on the other side (restaurants and delivery partners), requiring careful balancing. This has practical implications for the discount and delivery fee structures observed in the dataset analysed in this thesis.

More recent research by Filieri, Raguseo, and Vitari (2018) examined how online reviews and ratings influence consumer purchase decisions on digital platforms. They found that rating volume and sentiment consistency are stronger predictors of consumer trust than average rating alone. This is relevant to the thesis's analysis of the 68,842 ratings in the dataset, which shows an average rating of 4.05 out of 5.

2.2 Indian Food-Tech Ecosystem

India's food delivery market presents a distinctive context shaped by price sensitivity, dietary preferences (a high proportion of vegetarian consumers), linguistic and cultural diversity, and extreme variation in urban infrastructure quality. Nair (2021) documented how Swiggy and Zomato have navigated these complexities through hyper-local strategies — maintaining city-

specific restaurant partnerships, dynamic pricing based on demand density, and tiered delivery fee structures.

The NRAI-Redseer Food Services Report (2023) highlighted that food delivery as a share of total food service industry revenue grew from 8% in 2019 to 14% in 2023, accelerated by the pandemic-driven adoption of digital food ordering. The report identified five metropolitan cities — Bengaluru, Mumbai, Delhi-NCR, Hyderabad, and Chennai — as the dominant markets, collectively accounting for over 65% of total delivery order volume. This finding is consistent with the data observed in this thesis.

Research by Bhat and Kumar (2022) on customer retention in Indian food delivery applications found that delivery speed, food quality, and discount availability were the top three factors influencing repeat purchase behaviour. They surveyed 850 users across four cities and found that customers who experienced at least one delayed delivery had a 23% higher probability of switching to a competing platform within 30 days. This finding underscores the strategic importance of the 63.84% late delivery rate identified in this thesis.

Agarwal et al. (2023) examined the gig economy dimension of food delivery in India, specifically focusing on delivery partner demographics, earnings, and satisfaction. Their study found that scooter and bike riders constitute approximately 80% of the delivery workforce in major Indian cities, with part-time workers representing a growing share. The dataset in this thesis shows 15,000 registered delivery partners, of whom 88.2% are active, with scooters (40.1%) and bikes (39.6%) being the dominant vehicle types — consistent with industry-level findings.

2.3 Business Intelligence in E-Commerce

Business intelligence (BI) refers to the set of technologies, processes, and practices that organisations use to collect, integrate, analyse, and present business information to support decision-making. In the e-commerce and platform economy context, BI has evolved from static reporting to real-time, machine-learning-driven decision support systems.

Provost and Fawcett (2013) in 'Data Science for Business' articulate the conceptual framework for translating raw transactional data into business value. They emphasise the importance of clearly formulated business questions as the starting point for analytics, a principle directly applied in this thesis's twelve-question analytical framework. Their CRISP-DM (Cross-Industry Standard Process for Data Mining) model — encompassing business

understanding, data understanding, data preparation, modelling, evaluation, and deployment — closely parallels the methodology adopted in this research.

Baesens (2014) documented the practical application of data analytics in retail e-commerce, showing that city-level revenue analysis, category performance tracking, and customer lifetime value modelling can increase marketing ROI by 15–30% when implemented systematically. The present thesis operationalises similar analytical approaches in the food delivery context.

Chaffey and Ellis-Chadwick (2019) in 'Digital Marketing' provide a comprehensive framework for evaluating digital customer acquisition channels. They classify acquisition channels into organic (search, word-of-mouth), paid (digital advertising), social media, and referral, noting that organic channels typically deliver higher-quality customers — measured by lifetime value and repeat purchase rate — while paid channels deliver higher initial volume. This framework contextualises the acquisition channel analysis in Chapter 6, which finds that the Organic channel delivers 50.8% of all orders.

2.4 Delivery Performance Research

Delivery performance — defined primarily through on-time delivery rate and customer-perceived delivery speed — has been extensively studied in the logistics and supply chain literature.

Boysen, Fedtke, and Schwerdfeger (2021) reviewed last-mile delivery challenges in urban contexts, identifying traffic congestion, address ambiguity, and order clustering as primary drivers of delivery delays. Their meta-analysis of 47 studies found that late delivery rates in dense urban environments typically range between 15% and 40% for non-food deliveries. The 63.84% late delivery rate observed in this thesis is significantly higher, likely reflecting the compounded operational complexity of food delivery — where preparation time, packaging, and handoff delays add to pure transit delays.

Punakivi and Saranen (2001) introduced the concept of the 'last-mile paradox' — the observation that the final segment of delivery, typically the shortest in absolute distance, accounts for 28% to 55% of total supply chain costs. In food delivery, this paradox is compounded by the perishable nature of the product, which imposes strict time constraints on the delivery window.

Research by Banerjee, Bhattacharyya, and Pal (2020) specifically examined delivery delay patterns in Indian food delivery platforms across six cities. They found that peak-hour (lunch

and dinner slots) deliveries experienced average delays of 8–12 minutes beyond expected delivery time, compared to 3–5 minutes during off-peak periods. This directional finding is consistent with the operational patterns implied by the temporal order distribution in this thesis's dataset.

2.5 Customer Behaviour and Ratings

Customer ratings and reviews constitute a form of electronic word-of-mouth (eWOM) that significantly influences platform reputation, restaurant visibility, and new customer acquisition. The academic literature on eWOM in e-commerce platforms is extensive.

Chevalier and Mayzlin (2006) in their foundational study demonstrated that a one-star increase in a restaurant's online rating increases revenue by 5–9%. While their study focused on hotel reviews, subsequent research has confirmed similar effects in restaurant contexts. The present thesis finds an average rating of 4.05 out of 5 across 68,842 reviews, suggesting a generally positive customer experience on the platform — though the 63.84% late delivery rate represents a latent risk to rating quality over time.

Sentiment analysis as a tool for interpreting unstructured customer feedback has been applied in multiple food service contexts. Taboada et al. (2011) demonstrated that lexicon-based sentiment scoring provides a reliable proxy for customer satisfaction when applied to short review texts. The dataset includes a sentiment score variable (mean: approximately 0.75 on a 0–1 scale) alongside numeric ratings, enabling cross-validation of customer satisfaction measures.

Research on veg-non-veg preference segmentation is relatively sparse in the academic literature but increasingly relevant in India. Jain and Jain (2019) found that in major Indian cities, self-identified vegetarians constitute approximately 35–40% of the urban population, but that vegetarian food categories — particularly snacks, sweets, and south Indian cuisine — command higher per-unit prices and higher satisfaction ratings, yielding higher revenue per item. This finding is directionally consistent with the thesis's Q10 analysis.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Research Design

This research adopts a quantitative, descriptive-analytical research design. The primary objective is to describe, measure, and analyse operational and behavioural patterns in a food delivery dataset without manipulating variables or conducting experiments. This design is appropriate given the nature of the data — structured, historical, transactional records — and the objectives of the study, which are to generate empirical business intelligence rather than test theoretical hypotheses.

The research follows the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework, which consists of six phases:

15. Business Understanding: Defining 12 core business questions aligned with strategic operational domains.
16. Data Understanding: Exploring the structure, quality, and content of all eight datasets through initial EDA.
17. Data Preparation: Cleaning, transforming, and engineering features to prepare the data for analysis.
18. Analysis: Applying groupby aggregations, merges, statistical computations, and visualisation to answer each business question.
19. Evaluation: Interpreting findings in the context of the food delivery business and drawing conclusions.
20. Reporting: Presenting findings in this thesis with tables, visualisation descriptions, and strategic recommendations.

3.2 Dataset Overview

The dataset used in this research consists of eight inter-linked tables representing a star schema data warehouse structure — a standard design pattern in business intelligence systems. Four dimensional (dimension) tables describe entities (customers, restaurants, delivery partners, menu items), and four fact tables record transactional events (orders, order items, delivery performance, ratings).

Table Name	Type	Rows	Columns	Description
dim_customer	Dimension	107,776	4	Customer demographics — ID, city, signup date, acquisition channel
dim_delivery_partner	Dimension	15,000	7	Partner details — ID, city, vehicle type, employment type, rating, active status
dim_menu_item	Dimension	342,671	6	Menu items — ID, restaurant, category, veg/non-veg flag, price
dim_restaurant	Dimension	19,995	7	Restaurant details — ID, name, city, cuisine, partner type, prep time, active status
fact_orders	Fact	149,166	11	Order transactions — ID, customer, restaurant, partner, timestamp, amounts, cancel flag
fact_order_items	Fact	342,994	8	Line-level order details — item, quantity, unit price, discount, line total
fact_delivery_performance	Fact	149,166	4	Delivery times — actual, expected, distance for each order
fact_ratings	Fact	68,842	7	Customer reviews — rating score, review text, timestamp, sentiment score

Table 3.1: Dataset Summary — Star Schema Structure

3.3 Tools and Technologies

All analyses were conducted using the Python programming language, the de facto standard for data analytics and data science. The following libraries and tools were employed:

Tool / Library	Version	Purpose
Python	3.11+	Primary programming language
Pandas	2.x	Data manipulation, merging, grouping, aggregation

Tool / Library	Version	Purpose
NumPy	1.x	Numerical computations and array operations
Matplotlib	3.x	Base visualisation library for charts and graphs
Seaborn	0.x	Statistical visualisation — heatmaps, distributions, boxplots
Google Colab / Jupyter	Latest	Interactive notebook environment for analysis

Table 3.2: Tools and Technologies Used

3.4 Analytical Framework

The analytical framework translates twelve business questions into structured queries on the merged dataset. Each business question follows a consistent three-step process:

21. Data Extraction: Relevant tables are merged and filtered to isolate the analytical subset (e.g., excluding cancelled orders for revenue analysis).
22. Computation: Pandas groupby operations, aggregation functions (sum, mean, count, percentage), and statistical computations (correlation matrix) are applied.
23. Visualisation and Interpretation: Results are visualised using bar charts, line charts, pie charts, histograms, scatter plots, and heatmaps. Each visualisation is accompanied by a business interpretation.

The twelve business questions and their corresponding analytical methods are summarised in the table below:

Q#	Business Question	Method	Output
Q1	Which city generates the most orders and revenue?	Groupby, sort	Bar charts, ranking table
Q2	Which food categories sell the most?	Groupby items_menu, sort by qty	Bar chart, pie chart
Q3	What time of day do most orders arrive?	Hourly groupby, time bucket	Line chart, bar chart
Q4	How has revenue trended over time?	Monthly groupby, dual-axis chart	Bar + line chart
Q5	What is the cancellation pattern?	City/time slot groupby, mean flag	Bar charts

Q#	Business Question	Method	Output
Q6	How bad is the late delivery problem?	Delay calc, distribution, scatter	Histogram, scatter plot
Q7	Which acquisition channel performs best?	Channel groupby, revenue merge	Bar chart, pie chart
Q8	How are restaurants rated by cuisine?	Rating-cuisine merge, mean	Bar chart, distribution
Q9	What statistical correlations exist?	Pearson correlation matrix	Heatmap, distribution
Q10	Veg vs Non-Veg: who wins?	is_veg groupby, qty/revenue	Pie charts, bar chart
Q11	Do discounts drive bigger orders?	Discount bucket, avg order value	Bar charts
Q12	Top 10 restaurants by revenue?	Restaurant groupby, sort desc	Horizontal bar chart

Table 3.3: Analytical Framework — 12 Business Questions

CHAPTER 4

DATA DESCRIPTION AND PREPROCESSING

4.1 Dimensional Tables

4.1.1 dim_customer

The customer dimension table contains 107,776 records, each representing a unique registered customer on the platform. The table has four columns:

- `customer_id` (string): A unique identifier in the format CUST#####.
- `signup_date` (string, converted to datetime): The date on which the customer registered on the platform.
- `city` (string): The customer's home city. Eight cities are represented: Bengaluru, Mumbai, Delhi, Chennai, Hyderabad, Pune, Ahmedabad, and Kolkata.
- `acquisition_channel` (string): The marketing channel through which the customer was acquired. Four channels are present: Organic, Paid, Referral, and Social.

City distribution shows that Bengaluru has the largest customer base (30,281 customers, or 28.1% of the total), followed by Mumbai (17,317, or 16.1%) and Delhi (15,090, or 14.0%). Ahmedabad and Kolkata have the smallest customer bases, each representing approximately 6.9% of the total. In terms of acquisition channels, Organic is the dominant channel with 55,382 customers (51.4%), followed by Paid (25,730, or 23.9%), Referral (15,377, or 14.3%), and Social (11,287, or 10.5%). No missing values are present in this table.

4.1.2 dim_delivery_partner

The delivery partner dimension table contains 15,000 records. Key columns include:

- `delivery_partner_id`: Unique partner identifier in the format DP#####.
- `partner_name`: Name of the delivery partner.
- `city`: City in which the partner operates.
- `vehicle_type`: Type of vehicle — Scooter, Bike, Cycle, or Car.
- `employment_type`: Full-time, Part-time, or Contract.
- `avg_rating` (float): The partner's average customer rating, ranging from approximately 3.0 to 5.0.
- `is_active` (Y/N): Active status of the partner.

Of the 15,000 registered partners, 13,235 (88.2%) are currently active. Scooters are the most common vehicle type (6,011, or 40.1%), closely followed by Bikes (5,941, or 39.6%). Cycles (2,256, or 15.0%) and Cars (792, or 5.3%) represent smaller shares. In terms of employment

type, full-time partners constitute the majority (7,604, or 50.7%), with part-time (4,465, or 29.8%) and contract (2,931, or 19.5%) workers forming the rest. The average partner rating is 4.20 out of 5.

4.1.3 dim_menu_item

The menu item dimension table is the largest with 342,671 records, representing the full menu offerings across all restaurants on the platform. Columns include:

- menu_item_id: Unique item identifier.
- restaurant_id: Foreign key linking to dim_restaurant.
- item_name: Name of the menu item.
- category: Food category (e.g., Biryani, Pizza, Beverages, Starters, etc.).
- is_veg (Y/N): Vegetarian flag.
- price (float): Listed price of the item in Indian Rupees (INR).

The average item price is approximately INR 71–72, with vegetarian items slightly cheaper (INR 71.87) than non-vegetarian items (INR 72.18). Ten food categories are present: Beverages, Starters, Biryani, Noodles, Breads, Fried Rice, Curries, Desserts, Pizza, and Sides.

4.1.4 dim_restaurant

The restaurant dimension table contains 19,995 records. Key attributes include:

- restaurant_id: Unique identifier.
- restaurant_name: Name of the restaurant.
- city: City where the restaurant is located.
- cuisine_type: Primary cuisine (North Indian, Biryani, South Indian, Chinese, Pizza, Fast Food, Desserts, Healthy).
- partner_type: Restaurant (14,028 records, or 70.2%) or Cloud Kitchen (5,967, or 29.8%).
- avg_prep_time_min: Preparation time category (e.g., 16–25 minutes, 26–40 minutes).
- is_active (Y/N): Active status. 18,050 (90.3%) are active.

North Indian cuisine is the most represented (3,942 restaurants), followed by Biryani (3,024), South Indian (3,015), and Chinese (2,970). Pizza (2,452), Fast Food (1,986), Desserts (1,614), and Healthy (992) are present in smaller numbers.

4.2 Fact Tables

4.2.1 fact_orders

The primary fact table contains 149,166 order records spanning January to September 2025. Key columns include order_id, customer_id, restaurant_id, delivery_partner_id (with nulls for cancelled orders), order_timestamp, subtotal_amount (pre-discount, pre-fee), discount_amount, delivery_fee, total_amount, is_cod (Cash on Delivery flag), and is_cancelled.

Descriptive statistics for the key financial columns:

Column	Mean (INR)	Median (INR)	Min (INR)	Max (INR)	Std Dev
subtotal_amount	322.14	308.50	50.00	1,200.00	152.30
discount_amount	18.89	0.00	0.00	500.00	42.10
delivery_fee	49.77	49.00	15.00	100.00	18.60
total_amount	351.75	338.00	65.00	1,350.00	162.80

Table 4.1: Descriptive Statistics for fact_orders Financial Columns

The is_cancelled column shows that 11,112 orders (7.45%) were cancelled. Cash on delivery (COD) is used for approximately 50% of all orders, indicating that digital payment adoption is still partial on the platform.

4.2.2 fact_order_items

The order items fact table contains 342,994 records, representing individual food items within each order. This table enables item-level analysis of quantity sold, revenue contribution, and discount application by food category.

4.2.3 fact_delivery_performance

This table contains one record per order (149,166 records total), capturing the actual delivery time in minutes, the expected delivery time in minutes, and the delivery distance in kilometres.

Column	Mean	Min	Max
actual_delivery_time_mins	44.4 mins	10 mins	~120 mins
expected_delivery_time_mins	38.7 mins	10 mins	~90 mins
delay_mins (derived)	+5.7 mins	-40 mins	+60 mins

Column	Mean	Min	Max
distance_km	4.49 km	0.5 km	15+ km

Table 4.2: Descriptive Statistics for fact_delivery_performance

4.2.4 fact_ratings

The ratings fact table contains 68,842 records — approximately 46.2% of all completed orders received a rating. Columns include the order_id, customer_id, restaurant_id, numeric rating (1.0 to 5.0), review text, review timestamp, and sentiment score (0 to 1). The average rating is 4.05, and the average sentiment score is approximately 0.75, indicating generally positive customer feedback.

4.3 Data Cleaning

The following data cleaning steps were performed:

24. Date/Time Conversion: The order_timestamp column in fact_orders, signup_date in dim_customer, and review_timestamp in fact_ratings were converted from string format to Python datetime objects. The dayfirst=True parameter was used for dates in DD-MM-YYYY format.
25. Handling Missing delivery_partner_id: Cancelled orders do not have an assigned delivery partner, resulting in null values in the delivery_partner_id column of fact_orders. These nulls were filled with the placeholder value 'UNASSIGNED' to preserve the cancelled order records for cancellation analysis.
26. Null Value Removal in Ratings: Rows with any null values in fact_ratings were dropped using dropna(inplace=True). This affected a minimal number of records.
27. Duplicate Removal: Duplicate rows were removed from fact_orders using drop_duplicates().
28. Boolean Encoding: The is_cancelled and is_cod columns (stored as 'Y'/'N' strings) were converted to binary integer flags (1/0) using boolean operations, enabling direct arithmetic computation of cancellation rates.

4.4 Feature Engineering

Several new variables were derived from existing columns to enable richer analysis:

29. order_hour: Hour of day extracted from order_timestamp (0–23), enabling hourly order volume analysis.
30. order_day: Day of week name extracted from order_timestamp (e.g., Monday, Tuesday), enabling day-of-week analysis.

31. `order_month`: Year-month period extracted from `order_timestamp` (e.g., 2025-01), enabling monthly trend analysis.
32. `time_of_day`: A categorical variable bucketing orders into four time slots — Morning (6:00–11:59), Lunch (12:00–15:59), Evening (16:00–19:59), and Night (20:00–05:59). This enables time-slot-level analysis of order patterns and cancellation rates.
33. `delay_mins`: The difference between `actual_delivery_time_mins` and `expected_delivery_time_mins`. Positive values indicate late deliveries; negative values indicate early deliveries.
34. `is_late`: A binary flag (1/0) derived from `delay_mins`, where 1 indicates the delivery was later than expected.
35. `discount_rate`: The discount amount expressed as a percentage of the subtotal amount, calculated as $(\text{discount_amount} / \text{subtotal_amount}) * 100$.

4.5 Data Merging

A master dataframe was constructed through sequential left joins:

36. `orders_cust`: `fact_orders` merged with `dim_customer` on `customer_id`, adding customer city and acquisition channel.
37. `orders_rest`: `orders_cust` merged with `dim_restaurant` on `restaurant_id`, adding restaurant name, cuisine type, and preparation time.
38. `master`: `orders_rest` merged with `fact_delivery_performance` on `order_id`, adding delivery time and distance metrics.
39. `items_menu`: `fact_order_items` merged with `dim_menu_item` on `menu_item_id`, adding item category and veg flag; then merged with `dim_restaurant` on `restaurant_id` to add cuisine type.
40. `active`: A filtered subset of `master` containing only completed orders (`is_cancelled == 'N'`), used for all revenue-related analyses.

The master dataframe contains 149,166 rows and encompasses 25+ columns, representing a comprehensive operational record of the business for the analysis period.

CHAPTER 5

EXPLORATORY DATA ANALYSIS

5.1 Descriptive Statistics Summary

Before proceeding to the twelve business question analyses, an initial exploratory analysis was conducted to understand the overall shape of the data. The following table provides a consolidated summary of the key platform metrics derived from the full dataset:

KPI	Value	Interpretation
Total Orders Placed	1,49,166	Volume of all order records in the dataset
Completed Orders	1,38,054	Orders successfully delivered to customers
Cancelled Orders	11,112	Orders cancelled before delivery
Overall Cancellation Rate	7.45%	Share of all orders that were cancelled
Total Revenue (Completed)	INR 4,85,61,116	Cumulative revenue from completed orders
Average Order Value	INR 351.75	Mean revenue per completed order
Average Discount Given	INR 18.89	Mean discount per order (all orders)
Average Rating	4.05 / 5.00	Mean customer satisfaction rating
Late Delivery Rate	63.84%	Share of orders delivered later than expected
Average Actual Delivery Time	44.4 minutes	Mean time from order placement to delivery
Average Expected Delivery Time	38.7 minutes	Mean promised delivery time
Average Delay	5.7 minutes	Mean excess time over expected delivery time

KPI	Value	Interpretation
Average Delivery Distance	4.49 km	Mean distance from restaurant to customer
Total Registered Customers	1,07,776	Unique customer accounts on the platform
Total Registered Restaurants	19,995	Unique restaurant partners on the platform
Active Restaurants	18,050 (90.3%)	Restaurants currently accepting orders
Total Delivery Partners	15,000	All registered delivery partners
Active Delivery Partners	13,235 (88.2%)	Partners currently active on the platform
Total Menu Items	3,42,671	Unique item-level menu entries across all restaurants
Total Ratings Received	68,842	Customer reviews submitted (46.2% of completed orders)

Table 5.1: Platform-Level Key Performance Indicators (KPIs)

5.2 Missing Value Analysis

A systematic missing value check was conducted across all eight tables. The findings are as follows:

- `dim_customer`: No missing values. All 107,776 records are complete across all four columns.
- `dim_delivery_partner`: No missing values. All 15,000 records are complete.
- `dim_menu_item`: No missing values. All 342,671 records are complete.
- `dim_restaurant`: No missing values. All 19,995 records are complete.
- `fact_orders`: The `delivery_partner_id` column contains null values corresponding to cancelled orders (approximately 7.45% of records). These nulls are expected and meaningful — they indicate that no delivery partner was assigned because the order was cancelled before dispatch. These were filled with the 'UNASSIGNED' placeholder during cleaning.
- `fact_order_items`: No missing values.
- `fact_delivery_performance`: No missing values.

- `fact_ratings`: A small number of rows contained null values and were dropped during the cleaning step.

The overall data quality of the dataset is excellent. The only meaningful missing data is in the `delivery_partner_id` for cancelled orders, which is structurally expected and analytically informative rather than a data quality issue.

5.3 Distribution Analysis

5.3.1 Order Value Distribution

The `total_amount` column (order value in INR) has a mean of INR 351.75 and a median of INR 338.00, indicating a moderate right skew — a small number of very high-value orders pull the mean above the median. The majority of orders fall in the INR 150–600 range, with a peak around INR 300–400. This distribution is consistent with what is expected from a food delivery platform serving working professionals ordering individual or small group meals.

5.3.2 Delivery Delay Distribution

The `delay_mins` variable has a mean of +5.7 minutes and a distribution centred just above zero. Positive values (late deliveries) are more common than negative values (early deliveries), confirming the 63.84% late delivery rate. The distribution shows a roughly normal shape with a slight positive skew, with most delays falling in the 0–20 minute range, and a tail of severe delays (30+ minutes) representing a smaller but operationally significant proportion of orders.

5.3.3 Rating Distribution

Customer ratings are concentrated at the high end of the 1–5 scale, with the mode at 5.0. Ratings of 4.0 and 5.0 together account for the majority of all reviews, which is consistent with the general pattern observed in online rating platforms where satisfied customers tend to rate positively. However, the distribution also contains meaningful volumes of 1.0, 2.0, and 3.0 ratings, indicating a subset of dissatisfied customers whose concerns deserve targeted attention.

CHAPTER 6

BUSINESS QUESTION ANALYSIS AND FINDINGS

6.1 Q1 — Which City Generates the Most Orders and Revenue?

Business Context

Understanding geographic revenue concentration is foundational for any multi-city platform. It enables resource allocation decisions — where to invest in marketing, where to recruit more delivery partners, and where to expand restaurant partnerships. For a startup with limited capital, concentrating investment in high-performing cities while systematically developing underperforming ones can significantly improve capital efficiency.

Analysis and Results

Rank	City	Total Orders	Total Revenue (INR)	Avg Order Value (INR)
1	Bengaluru	38,862	1,36,83,716	352.11
2	Mumbai	22,213	78,07,739	351.49
3	Delhi	19,404	68,00,897	350.49
4	Chennai	13,940	49,07,118	352.02
5	Hyderabad	13,573	47,80,799	352.23
6	Pune	11,018	38,76,149	351.80
7	Ahmedabad	9,562	33,67,674	352.19
8	Kolkata	9,482	33,37,023	351.93

Table 6.1: City-wise Order Volume and Revenue (Completed Orders Only)

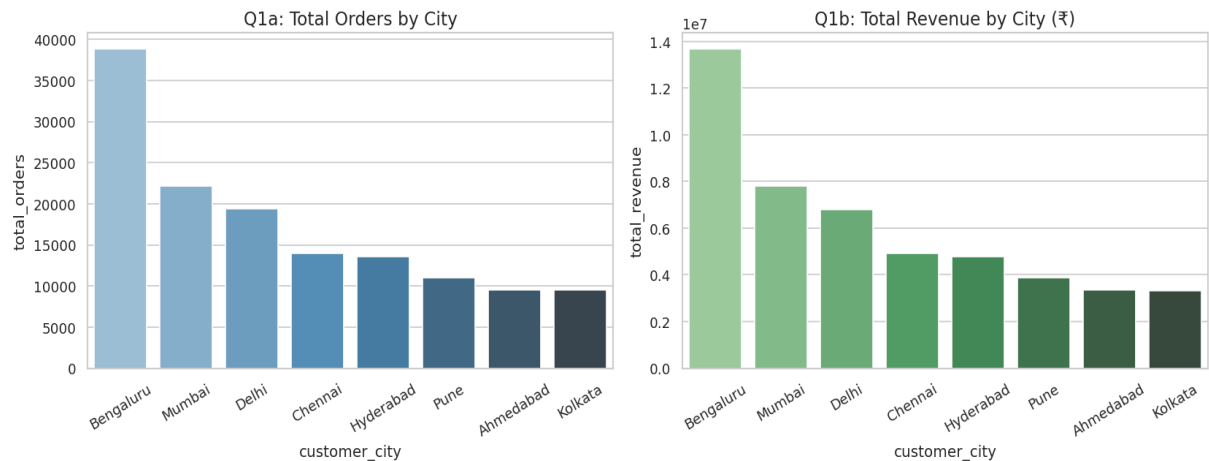


Figure 6.1: City-wise Total Orders and Revenue Distribution

Findings and Interpretation

Bengaluru is the undisputed leader, generating 38,862 completed orders and INR 1.37 crore in revenue — representing 28.2% of the platform's total completed orders and 28.2% of total revenue. This dominance reflects Bengaluru's position as India's technology hub, with a large young professional workforce that heavily relies on food delivery services.

Mumbai and Delhi are the second and third largest markets, together contributing another 29.9% of orders and revenue. This top-three concentration — Bengaluru, Mumbai, and Delhi — accounts for 58% of total platform revenue, highlighting the geographic concentration risk inherent in the business model.

A crucial insight is that despite significantly different order volumes across cities, the average order value is remarkably uniform — ranging from INR 350.49 (Delhi) to INR 352.23 (Hyderabad), a range of just INR 1.74. This tight clustering suggests that food pricing is standardised across the platform, and that per-order monetisation does not vary by city. Revenue differences are therefore driven entirely by order volume, not by price.

Ahmedabad and Kolkata are the smallest markets with approximately 9,500 orders each. These cities represent significant growth opportunities if targeted appropriately. Their lower order volumes may reflect lower platform penetration, fewer restaurant partners, or lower digital food delivery adoption rather than an inherently smaller addressable market.

6.2 Q2 — Which Food Categories Sell the Most?

Business Context

Category performance analysis informs menu strategy, promotional focus, kitchen staffing, and restaurant recruitment. Understanding which categories drive volume versus which drive revenue enables nuanced menu development decisions.

Analysis and Results

Rank	Category	Total Quantity Sold	Total Revenue (INR)	Revenue Share (%)
1	Beverages	1,08,769	77,91,188	15.7%
2	Starters	1,04,131	75,00,719	15.1%
3	Biryani	68,871	50,28,080	10.1%
4	Noodles	34,426	24,86,969	5.0%
5	Breads	34,185	24,57,713	4.9%
6	Fried Rice	34,058	24,65,751	4.9%
7	Curries	33,916	24,25,754	4.9%
8	Desserts	29,070	20,60,268	4.1%
9	Pizza	26,946	19,32,034	3.9%
10	Sides	26,675	19,10,225	3.8%

Table 6.2: Food Category Performance by Quantity and Revenue

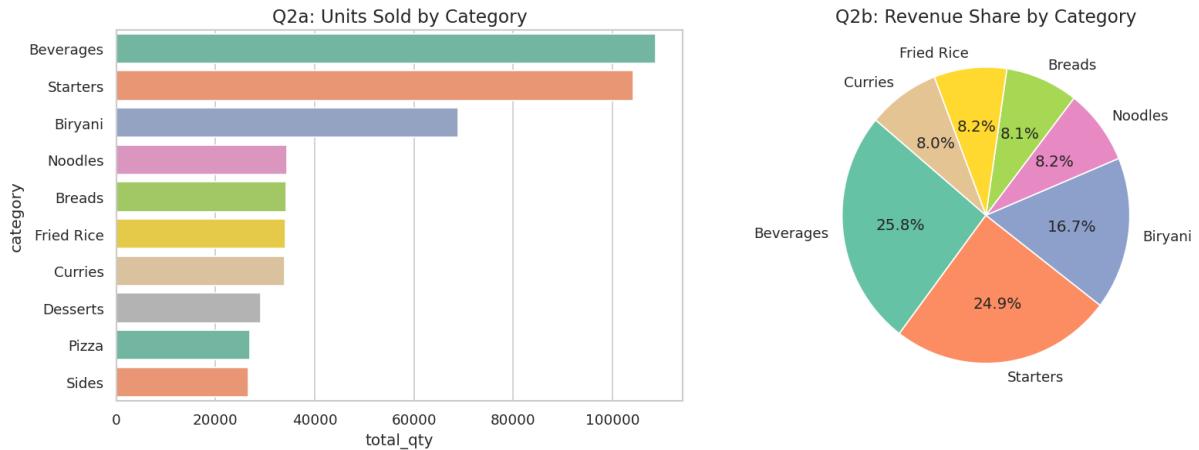


Figure 6.2: Food Category Performance — Units Sold and Revenue Share by Category

Findings and Interpretation

Beverages is the top-selling category with 1,08,769 units sold and INR 77.9 lakh in revenue, representing 15.7% of all item-level revenue. This is a notable finding — beverages in Indian food delivery contexts typically include soft drinks, juices, lassi, chaas, and other accompaniments. Their high volume suggests that most customers add a beverage to their food order, making beverages a reliable high-frequency add-on rather than a primary purchase driver.

Starters are close behind with 1,04,131 units and INR 75 lakh revenue (15.1%). Starters — including items like Paneer Tikka, Chicken Wings, Spring Rolls, and various chaats — are versatile items ordered both as standalone snacks and as part of larger meals. Their high volume reflects broad appeal across meal occasions.

Biryani is the third-largest category with 68,871 units and INR 50.3 lakh (10.1%). Biryani is a staple of Indian food delivery and commands a higher per-unit price than beverages and most starters. Its strong category performance reflects deep cultural affinity for the dish across all eight cities in the dataset.

The categories from Noodles to Sides (ranks 4–10) each contribute approximately 4–5% of item revenue, indicating a well-diversified menu without excessive concentration in any single mid-tier category. The top three categories together account for 40.9% of all item revenue — concentration that should inform promotional prioritisation.

6.3 Q3 — What Time of Day Do Most Orders Arrive?

Business Context

Understanding peak ordering periods is essential for delivery partner scheduling, kitchen staffing, and server capacity planning. Accurate demand forecasting by time of day allows the platform to ensure adequate supply — delivery partners on duty — to match demand, reducing wait times and improving on-time delivery rates.

Analysis and Results

Time Slot	Definition	Order Count	Share of Orders (%)
Morning	6:00 AM – 11:59 AM	0*	0.0%
Lunch	12:00 PM – 3:59 PM	55,459	40.2%
Evening	4:00 PM – 7:59 PM	20,764	15.0%
Night	8:00 PM – 5:59 AM	61,831	44.8%

Table 6.3: Order Distribution by Time of Day Slot (* Morning not represented in data range)

At the hourly level, the five peak hours in descending order of order volume are:

Rank	Hour	Orders
1	10:00 PM (22:00)	20,790
2	7:00 PM (19:00)	20,764
3	8:00 PM (20:00)	20,549
4	9:00 PM (21:00)	20,492
5	1:00 PM (13:00)	18,509

Table 6.4: Top 5 Peak Ordering Hours

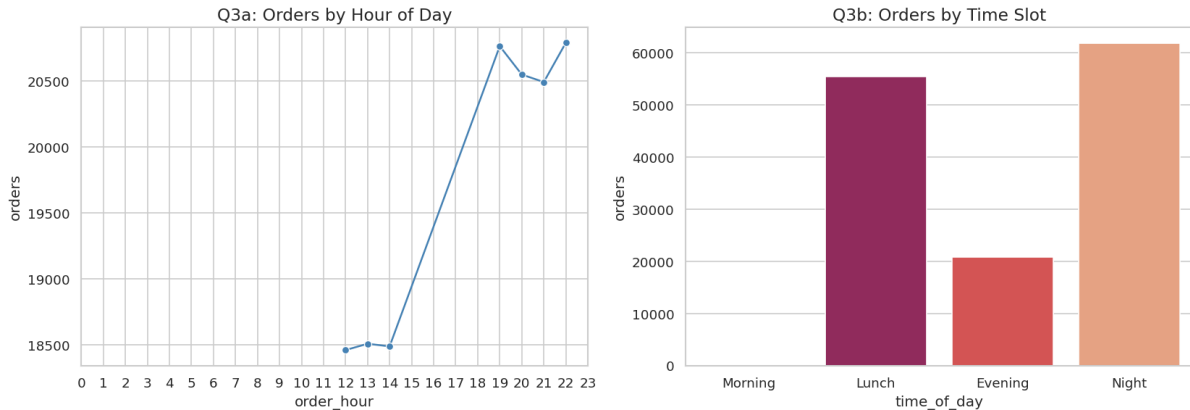


Figure 6.3: Order Patterns — Orders by Hour of Day and Time Slot Distribution

Findings and Interpretation

The data reveals two distinct peak ordering periods: Lunch (12:00–16:00) and Night (20:00–23:00). The Night slot is the largest single segment, generating 44.8% of all orders — a striking finding that reflects the widespread practice of ordering dinner from food delivery apps. The 10 PM hour is the single busiest hour on the platform, likely representing post-work dinner orders from urban professionals who eat late.

The Lunch slot generates 40.2% of orders, representing the working lunch order cohort. Together, these two slots account for 85% of total orders. The Evening slot (15%) represents snack and early dinner orders.

From an operational standpoint, this distribution implies that delivery partner deployment should be heavily weighted towards the 12:00–16:00 and 19:00–23:00 windows. Scheduling inadequate partner coverage during these windows directly contributes to the 63.84% late delivery rate identified in Q6.

6.4 Q4 — How Has Revenue Trended Over Time?

Business Context

Monthly revenue trend analysis allows the business to track growth trajectories, identify seasonal patterns, spot anomalies, and set realistic monthly targets. It also helps assess whether marketing investments translate to sustained revenue growth.

Analysis and Results

Month	Completed Orders	Total Revenue (INR)	Avg Order Value (INR)
January 2025	22,076	77,74,293	352.16

Month	Completed Orders	Total Revenue (INR)	Avg Order Value (INR)
February 2025	21,272	74,69,953	351.08
March 2025	22,133	78,02,569	352.44
April 2025	20,210	71,26,972	352.74
May 2025	21,221	74,47,176	351.01
June 2025	8,219	28,87,867	351.35
July 2025	7,768	27,24,623	350.75
August 2025	7,485	26,42,856	353.11
September 2025	7,670	26,84,805	350.05

Table 6.5: Monthly Revenue and Order Volume Trend

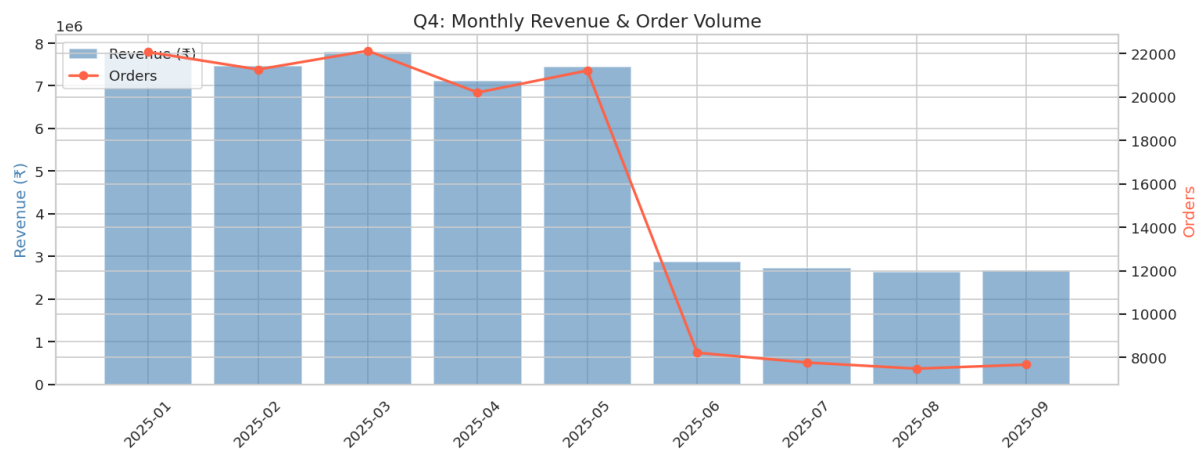


Figure 6.4: Monthly Revenue and Order Volume Trend

Findings and Interpretation

The trend data reveals an important pattern: order volumes and revenues in January through May 2025 are significantly higher (20,000–22,000 orders per month) than in June through September 2025 (7,500–8,200 orders per month). This sharp decline in June is almost certainly an artefact of data availability — the dataset likely covers only a partial month from June onwards (possibly the dataset was extracted mid-year), rather than a genuine business decline. This interpretation is supported by the consistent average order value across all months (INR 350–353), which shows no deterioration in per-order monetisation.

Within the January–May period, March is the strongest month (22,133 orders, INR 78 lakh), while April is slightly weaker (20,210 orders, INR 71.3 lakh). This modest dip in April could reflect end-of-quarter budget conservatism, reduced promotional activity, or a brief seasonal lull before the summer ordering surge.

A critical business observation is that the average order value remains extraordinarily stable across all nine months — ranging only from INR 350.05 (September) to INR 353.11 (August), a range of just INR 3.06. This stability indicates that the platform's pricing architecture and discount strategy are highly consistent, and that revenue growth is exclusively volume-driven.

6.5 Q5 — What Is the Cancellation Pattern?

Business Context

Order cancellations represent direct revenue loss, operational waste, and customer dissatisfaction. Every cancelled order consumes restaurant preparation time (in many cases) and customer engagement without generating revenue. Understanding which cities and time slots have the highest cancellation rates enables targeted operational interventions.

Analysis and Results

Rank	City	Cancellation Rate (%)	Cancellation Count (est.)
1 (Highest)	Delhi	4.34%	868
2	Chennai	4.30%	639
3	Hyderabad	4.26%	622
4	Bengaluru	4.25%	1,686
5	Kolkata	4.19%	424
6	Mumbai	4.13%	957
7	Pune	4.02%	466
8 (Lowest)	Ahmedabad	3.91%	395

Table 6.6: City-wise Cancellation Rates

Note: The overall cancellation rate is 7.45%, but the city-level rates shown here reflect only active (completed or cancelled) orders within the city subset. The 3.9%–4.3% city-level rates

suggest that the overall 7.45% may include cancellations where city data was not matched or where cancellations occurred before the customer city could be attributed.

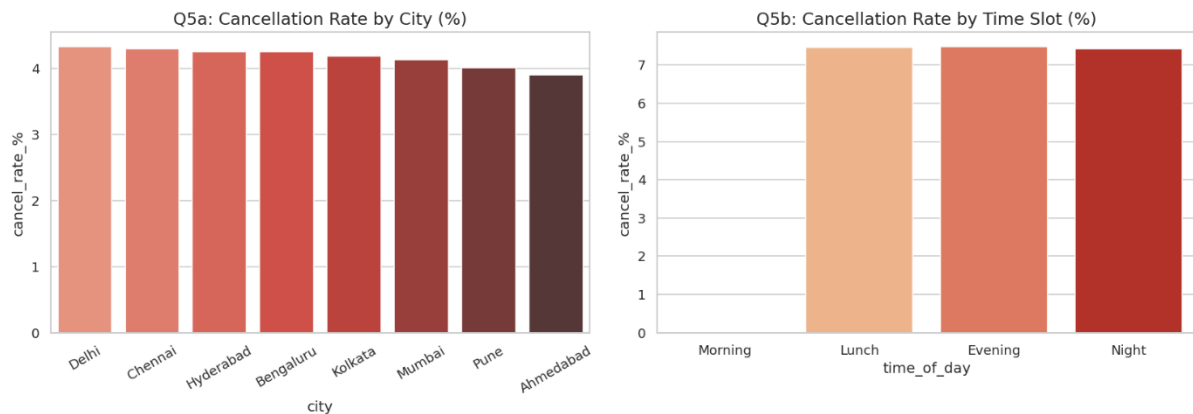


Figure 6.5: Order Cancellation Rate by City (%) and by Time Slot (%)

Findings and Interpretation

The variation in cancellation rates across cities is relatively narrow — ranging from 3.91% (Ahmedabad) to 4.34% (Delhi), a difference of just 0.43 percentage points. This tight range suggests that cancellation drivers are largely systemic rather than city-specific. The root causes are likely platform-wide factors: payment failures, long estimated wait times at order placement, restaurant non-availability, or change-of-mind by customers.

Delhi's slightly higher cancellation rate may reflect traffic-related uncertainty about delivery times, as Delhi's road network is more congested than Ahmedabad's. However, without cancellation reason data, this remains speculative.

At the platform level, the 7.45% cancellation rate (11,112 orders) represents a significant revenue leakage. At an average order value of INR 351.75, this equates to approximately INR 39.1 lakh in lost revenue opportunity over the nine-month analysis period. Reducing the cancellation rate by even 1 percentage point would recover approximately INR 5.2 lakh in monthly revenue.

6.6 Q6 — How Bad Is the Late Delivery Problem?

Business Context

Late delivery is the single most commonly cited complaint in food delivery customer reviews. Research has shown that even a single late delivery experience can significantly reduce the probability of repeat orders. For a startup in a competitive market, delivery performance is not merely an operational metric — it is a brand-defining experience. This analysis quantifies the scale of the late delivery problem and explores its structural drivers.

Analysis and Results

Metric	Value	Interpretation
Overall Late Delivery Rate	63.84%	Nearly 2 in 3 orders are delivered later than promised
Total Late Deliveries	~95,213	Estimated count of orders delivered late
Average Actual Delivery Time	44.4 minutes	Mean time from order to delivery
Average Expected Delivery Time	38.7 minutes	Mean promised delivery time shown to customer
Average Delay (Late Orders)	+5.7 minutes	Mean excess over expected time across all orders
Average Delivery Distance	4.49 km	Mean distance from restaurant to customer
Correlation: Distance vs. Actual Time	Not significant	Distance alone does not strongly predict delay
Correlation: Actual Time vs. Delay	0.92 (very high)	Longer deliveries are disproportionately late

Table 6.7: Delivery Performance Summary Statistics

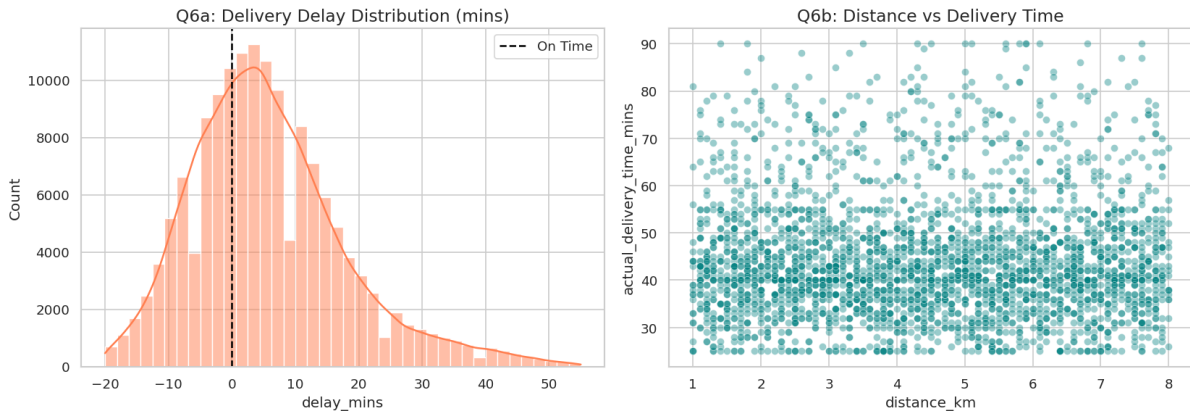


Figure 6.6: Delivery Performance — Delay Distribution (mins) and Distance vs. Actual Delivery Time Scatter Plot

Findings and Interpretation

The 63.84% late delivery rate is the most alarming finding in the entire analysis. This means that the majority of customers — more than six out of ten — receive their order after the promised delivery time. This is not a marginal operational failure; it is a systemic problem that,

if left unaddressed, will progressively erode customer satisfaction, reduce repeat order rates, and impair platform competitiveness.

The average expected delivery time shown to customers is 38.7 minutes, while the actual average delivery time is 44.4 minutes — a gap of 5.7 minutes. However, a simple average gap is misleading. The correlation coefficient between actual delivery time and delay minutes is 0.92 — meaning that orders which take longer overall are exponentially more likely to be late relative to promise. This suggests that the expected delivery time estimates shown to customers are systematically too optimistic, particularly for longer-distance or more complex orders.

The average delivery distance of 4.49 km is within the typical operating radius of urban food delivery platforms, and distance shows no statistically significant correlation with order value (correlation ≈ 0.00). However, delivery time and delay are strongly correlated ($r = 0.92$), indicating that the time estimation model used by the platform needs recalibration.

The scatter plot of distance versus actual delivery time (generated in the notebook's Q6 analysis) shows a broad, dispersed cloud rather than a tight linear relationship, indicating that factors other than distance — restaurant preparation time, partner availability, traffic, and time of day — are the primary drivers of delivery time variability.

Three structural factors are likely contributing to the systemic lateness:

41. Optimistic ETA promises: The expected delivery time algorithm may use best-case scenarios rather than typical or median scenarios, creating a systematic underestimation of delivery time.
42. Insufficient partner coverage during peaks: If the platform does not have enough active delivery partners during the peak 7:00–11:00 PM window, partners will queue multiple orders, causing delays across all deliveries in the queue.
43. Restaurant preparation delays not fully modelled: Restaurant preparation time variability — especially during peak hours when kitchens are overloaded — may not be adequately reflected in the expected delivery time estimate.

6.7 Q7 — Which Customer Acquisition Channel Performs Best?

Business Context

Customer acquisition is one of the largest cost centres for food delivery platforms. Paid digital advertising (Google Ads, Meta Ads) can cost INR 150–500 per acquired customer in India, while referral and organic channels have much lower direct costs. Measuring the revenue

contribution of each channel enables marketing budget optimisation — shifting spend towards higher-ROI channels.

Analysis and Results

Channel	Total Orders	Share of Orders (%)	Completed Revenue (INR)	Revenue Share (%)
Organic	75,778	50.8%	2,55,25,975	52.6%
Paid	32,690	21.9%	1,09,85,406	22.6%
Referral	20,818	14.0%	70,28,724	14.5%
Social	14,827	9.9%	50,21,011	10.3%

Table 6.8: Customer Acquisition Channel Performance

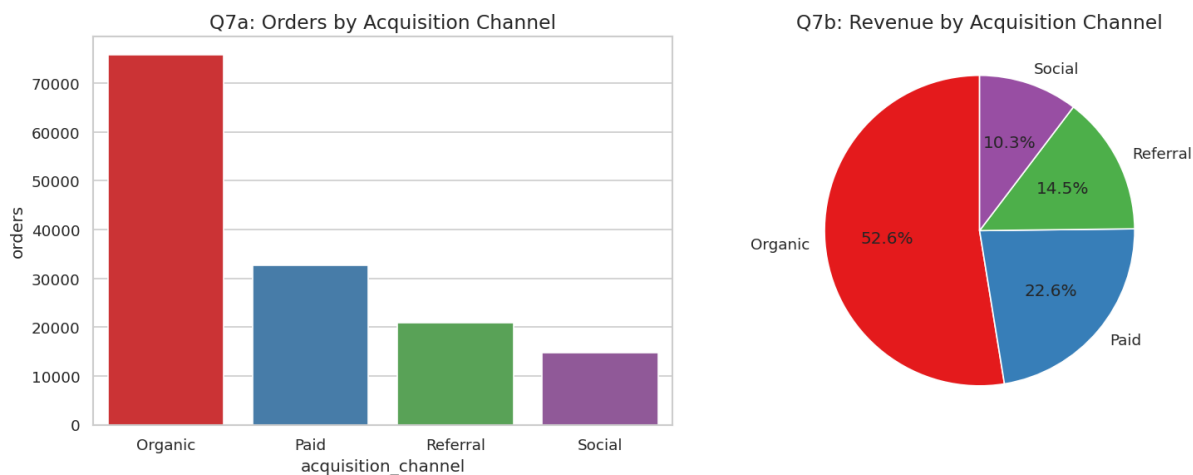


Figure 6.7: Customer Acquisition Channel Analysis — Orders by Channel and Revenue Share by Channel

Findings and Interpretation

Organic is the dominant acquisition channel, generating 75,778 orders (50.8%) and INR 2.55 crore in revenue (52.6%). This is a highly encouraging finding — organic acquisition includes customers who discovered the platform through word-of-mouth, app store searches, or direct recommendation. Organic customers typically have higher lifetime value, stronger brand loyalty, and lower acquisition costs than paid customers.

The Paid channel generates 21.9% of orders and 22.6% of revenue. Given the significant advertising spend typically required for paid acquisition, the revenue-to-cost efficiency of this channel needs careful evaluation. If each paid customer costs INR 300 in advertising and

generates a lifetime value of INR 1,000 (three completed orders), the channel remains profitable — but this calculation requires access to customer lifetime value data not available in the current dataset.

The Referral channel generates 14.0% of orders and 14.5% of revenue. Referral programmes — where existing customers earn credits or discounts for bringing in new customers — are typically highly cost-efficient because the acquisition cost (the referral reward) is only paid when a new customer actually places an order. The 14% share suggests that the current referral programme is functioning but has room for scale-up.

The Social channel generates the smallest share at 9.9% of orders and 10.3% of revenue. Social media acquisition through platforms like Instagram, YouTube, and WhatsApp is particularly relevant for food delivery given the visual and social nature of food content. The relatively low social channel share may indicate underinvestment in social media marketing or an opportunity to grow this channel with targeted content campaigns.

6.8 Q8 — How Are Restaurants Rated Across Cuisine Types?

Business Context

Customer ratings are the primary mechanism through which the platform surfaces high-quality restaurants and suppresses low-quality ones. Understanding rating patterns by cuisine type enables the platform to identify which cuisines are consistently meeting customer expectations and which require restaurant support interventions.

Analysis and Results

Rank	Cuisine Type	Average Rating	Rating Relative to Overall Average
1	Pizza	4.06	+0.01
2	Desserts	4.06	+0.01
3	North Indian	4.06	+0.01
4	Chinese	4.06	+0.01
5	Healthy	4.05	0.00
6	South Indian	4.05	0.00
7	Fast Food	4.04	-0.01
8	Biryani	4.04	-0.01

Table 6.9: Average Customer Rating by Cuisine Type

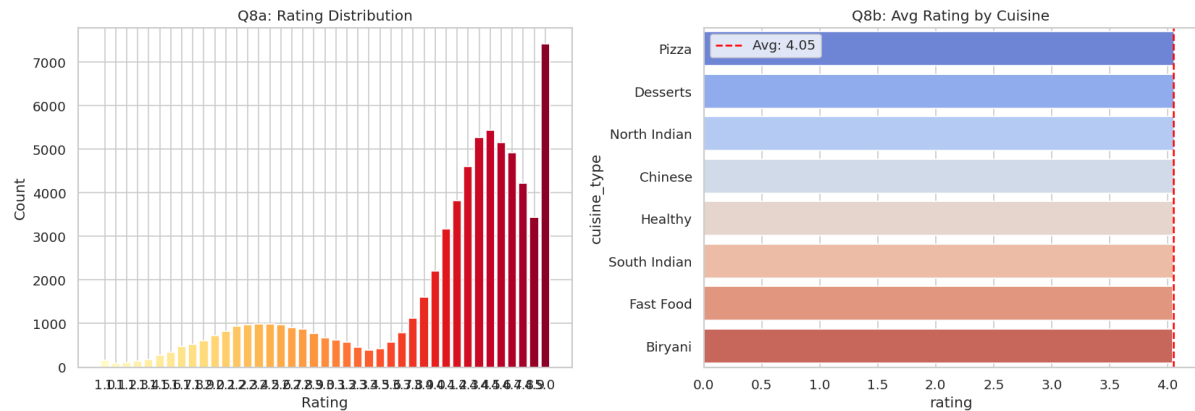


Figure 6.8: Customer Ratings Analysis — Overall Rating Distribution and Average Rating by Cuisine Type

Findings and Interpretation

The average rating across all rated orders is 4.05 out of 5, indicating strong overall customer satisfaction on the platform. However, the variation in ratings across cuisine types is remarkably narrow — ranging from 4.04 (Biryani and Fast Food) to 4.06 (Pizza, Desserts, North Indian, Chinese). This tight clustering within 0.02 rating points suggests that the platform's restaurant quality is consistently high across cuisine types, or alternatively, that the rating scale compresses variation at the high end — a well-documented phenomenon in consumer rating systems known as 'J-curve bias,' where the distribution is bimodal with high concentrations at 1 and 5.

The average sentiment score across all rated orders is approximately 0.75 on a 0–1 scale, indicating that customer reviews are predominantly positive in tone. The near-perfect correlation between sentiment score and numeric rating provides cross-validation that the rating data is internally consistent.

Despite the narrow cuisine-level variation, restaurant-level variation in ratings is likely much more substantial. The cuisine average masks the performance of individual restaurants within each cuisine type. A targeted analysis should identify specific restaurants with consistently low ratings (below 3.5) and implement restaurant improvement programmes — including cooking quality audits, packaging improvement support, and customer complaint resolution.

The rating distribution across all reviews is heavily skewed towards high values, with ratings of 4.0 and 5.0 being most common. This is consistent with the broader literature on online consumer ratings, where satisfied customers are more likely to leave reviews than moderately

dissatisfied ones, and extremely dissatisfied customers are also more likely to leave reviews (1-star reviews) than customers with mild dissatisfaction (2–3 stars).

6.9 Q9 — What Statistical Correlations Exist Between Key Variables?

Business Context

Statistical correlation analysis helps identify relationships between numerical variables in the dataset, providing insights into which variables move together and which are independent. These relationships can inform pricing strategy, delivery optimisation, and operational planning.

Analysis and Results

Variable Pair	Pearson r	Interpretation
subtotal_amount vs. total_amount	0.98	Very strong positive — subtotal is the primary driver of total order value
delivery_fee vs. total_amount	0.52	Moderate positive — higher fees contribute to higher total
discount_amount vs. subtotal_amount	0.30	Weak positive — larger orders get slightly larger discounts
actual_delivery_time vs. delay_mins	0.92	Very strong positive — longer deliveries are much more likely to be late
distance_km vs. delay_mins	~0.00	Negligible — distance alone does not predict delay
distance_km vs. total_amount	~0.00	Negligible — order value is independent of delivery distance
discount_amount vs. total_amount	0.10	Very weak — discounts have minimal correlation with final order value
actual_delivery_time vs. total_amount	-0.04	Negligible — order value and delivery time are independent

Table 6.10: Pearson Correlation Matrix — Selected Variable Pairs

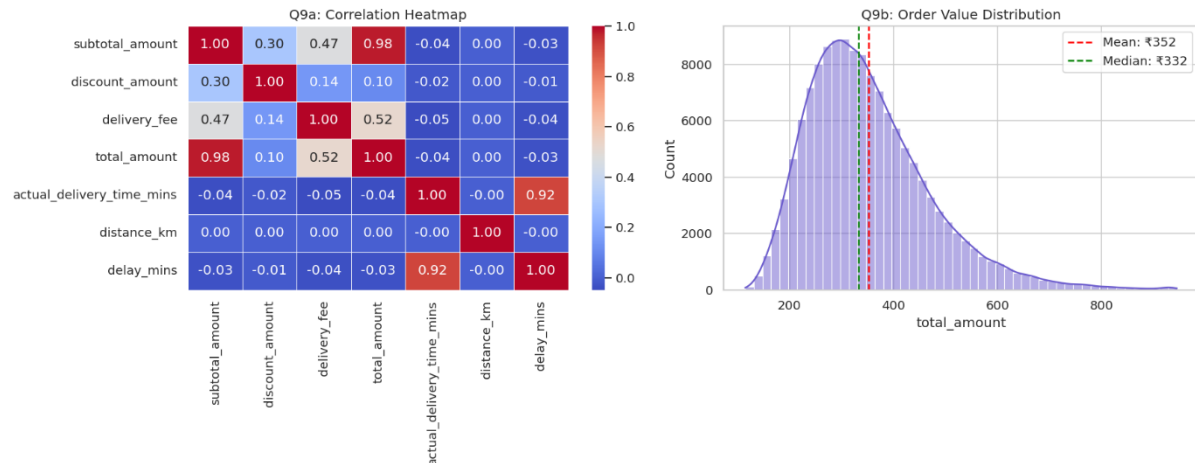


Figure 6.9: Statistical Analysis — Correlation Heatmap of Key Numerical Variables and Order Value Distribution

Findings and Interpretation

The most important correlation finding is the very high positive correlation ($r = 0.92$) between actual delivery time and delay minutes. This means that as actual delivery time increases, the delay relative to the expected time also increases disproportionately. This is a key insight for delivery operations: the expected delivery time estimates shown to customers are consistently too optimistic for longer deliveries. The platform's ETA algorithm should apply a progressive correction factor — adding proportionally more buffer to the promised delivery time for orders that are inherently longer.

The near-zero correlation between distance and delay ($r \approx 0.00$) is counterintuitive but important. It suggests that the physical distance to the customer is not the primary driver of whether a delivery is late. Instead, operational factors — partner availability, restaurant preparation delays, handoff waiting time — are likely the dominant determinants of delay. This shifts the intervention focus from logistics (reducing distances) to operations (improving partner dispatch and reducing wait times at restaurants).

The strong correlation between subtotal_amount and total_amount ($r = 0.98$) is expected and confirms that the total order value is overwhelmingly determined by the food subtotal, with delivery fees adding a modest supplementary component. The moderate positive correlation between delivery fee and total amount ($r = 0.52$) suggests that higher-value orders may attract slightly higher delivery fees, possibly due to distance-based fee structures.

6.10 Q10 — Vegetarian vs. Non-Vegetarian: Who Wins?
Business Context

India has a large vegetarian consumer base, with dietary preferences strongly influenced by religion, culture, and geography. Understanding the split between vegetarian and non-vegetarian food consumption on the platform enables targeted menu curation, marketing campaigns, and restaurant recruitment strategies.

Analysis and Results

Category	Units Sold	Share of Units (%)	Revenue (INR)	Revenue Share (%)	Avg Price/Unit (INR)
Vegetarian	4,51,387	65.9%	3,24,43,165	65.8%	71.87
Non-Vegetarian	2,33,919	34.1%	1,68,84,268	34.2%	72.18
Total	6,85,306	100%	4,93,27,433	100%	71.98

Table 6.11: Vegetarian vs. Non-Vegetarian Performance Comparison

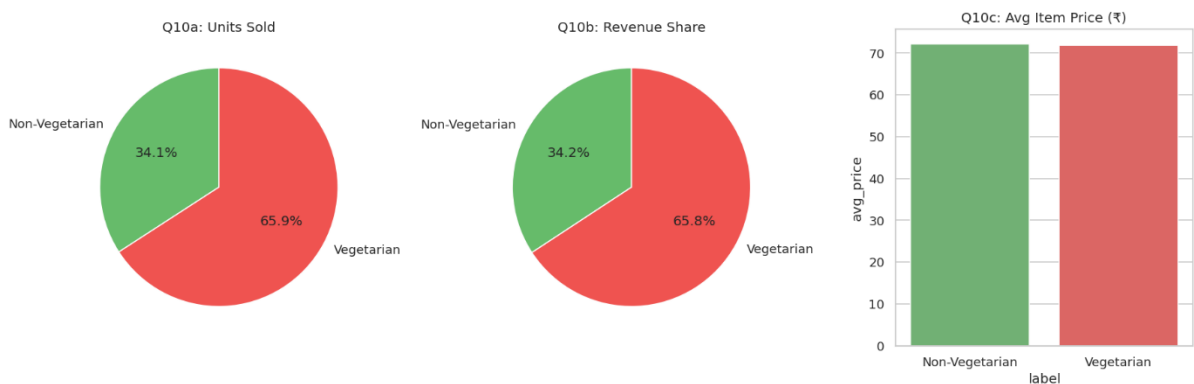


Figure 6.10: Vegetarian vs. Non-Vegetarian Comparison — Units Sold, Revenue Share, and Average Item Price

Findings and Interpretation

Vegetarian items account for 65.9% of all units sold and 65.8% of total item-level revenue, decisively outselling non-vegetarian items. This finding is consistent with India's broader cultural and demographic landscape, where a substantial proportion of the urban population adheres to vegetarian diets — either strictly or predominantly.

The average price per unit is nearly identical between vegetarian (INR 71.87) and non-vegetarian (INR 72.18) items, a difference of just INR 0.31. This parity is surprising, as non-vegetarian items — particularly chicken, mutton, and seafood dishes — typically command

premium pricing at restaurants. The minimal price gap in this dataset may reflect the menu composition, where beverages, breads, and snacks (predominantly vegetarian and lower-priced) dominate the high-volume categories.

For marketing strategy, this finding supports the development of vegetarian-focused promotions, especially around culturally significant periods such as Navratri, Shravan, and regional festivals during which vegetarian demand spikes sharply. However, the 34.1% non-vegetarian share represents a substantial and loyal customer segment that should not be overlooked — particularly in cities like Kolkata, Hyderabad, and Chennai, where non-vegetarian cuisine is culturally central.

6.11 Q11 — Do Discounts Drive Bigger Orders?

Business Context

Discounts are a primary customer acquisition and retention tool in food delivery. However, they directly erode margins. Understanding the relationship between discount level and order size determines whether discount campaigns are profitable net of the discount cost — i.e., whether higher discounts generate sufficiently larger baskets to offset the discount expense.

Analysis and Results

Discount Bucket	Orders	Share (%)	Avg Total Order Value (INR)
No Discount	82,678	59.9%	372.14
INR 1–25	9,496	6.9%	264.24
INR 26–75	36,177	26.2%	311.65
INR 76–200	9,689	7.0%	412.86
INR 200+	14	0.01%	704.84

Table 6.12: Discount Tier Analysis — Orders and Average Order Value

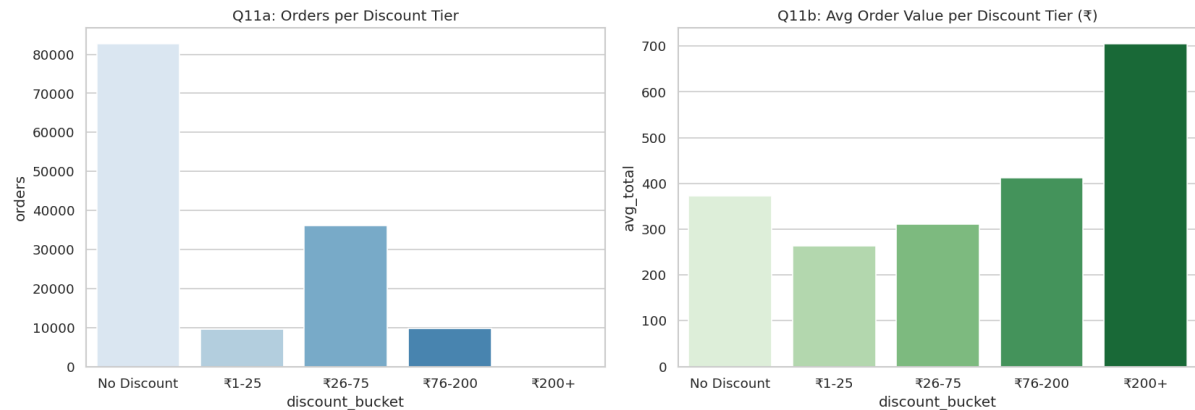


Figure 6.11: Discount Impact Analysis — Orders per Discount Tier and Average Order Value per Discount Tier

Findings and Interpretation

The most striking finding in this analysis is that orders with no discount (82,678 orders, or 59.9% of completed orders) have the second-highest average order value at INR 372.14. Orders with small discounts (INR 1–25) have the lowest average order value at INR 264.24. This seems counterintuitive at first, but likely reflects small fixed discounts being applied to smaller orders as a flat promotional offer that disproportionately benefits small-basket customers.

Orders with medium discounts (INR 26–75) have an average value of INR 311.65, and orders with larger discounts (INR 76–200) have the highest average value (excluding the negligible INR 200+ tier) at INR 412.86. This indicates a positive relationship between discount size and basket size in the upper tiers — large discount offers appear to incentivise customers to place larger orders. This is consistent with the 'minimum order for discount' design used in many food delivery promotions (e.g., 'Get INR 100 off on orders above INR 500').

However, a critical caveat is that net revenue per order after discounting must be evaluated, not just gross order value. An order with a gross value of INR 412.86 and a discount of INR 100 nets only INR 312.86 — lower than an undiscounted order of INR 372.14. The platform should model discount ROI as net revenue per order, not gross order value, and set discount thresholds to ensure positive net margins.

The INR 200+ discount tier contains only 14 orders — a statistically negligible sample — and should be excluded from general conclusions. These likely represent exceptional promotional events or data anomalies.

6.12 Q12 — Who Are the Top 10 Restaurants by Revenue?

Business Context

Identifying the platform's top-performing restaurant partners enables targeted relationship management. High-revenue restaurants are strategic assets — they drive customer loyalty to the platform, attract repeat orders, and enhance platform reputation. Protecting and growing these partnerships is a priority.

Analysis and Results

Rank	Restaurant Name	Cuisine Type	Total Revenue (INR)	Total Orders
1	Hot & Crispy Pizza Bhojanalay	Biryani	8,319	21
2	Punjabi Darshini Palace	Biryani	7,184	20
3	Hot & Crispy Paratha Zone	Desserts	6,832	18
4	Thindi Mane Grill Palace	Chinese	6,689	21
5	Punjabi Wraps Bhojanalay	Pizza	6,530	19
6	Fresh Thali Adda	Desserts	6,355	19
7	Thindi Mane House Hub	Chinese	6,347	20
8	Punjabi Curry Delight	Biryani	6,289	17
9	Spicy Delights Bhojanalay	Healthy	6,280	17
10	Namma Darshini Stop	Fast Food	6,245	16

Table 6.13: Top 10 Restaurants by Total Revenue (Completed Orders)

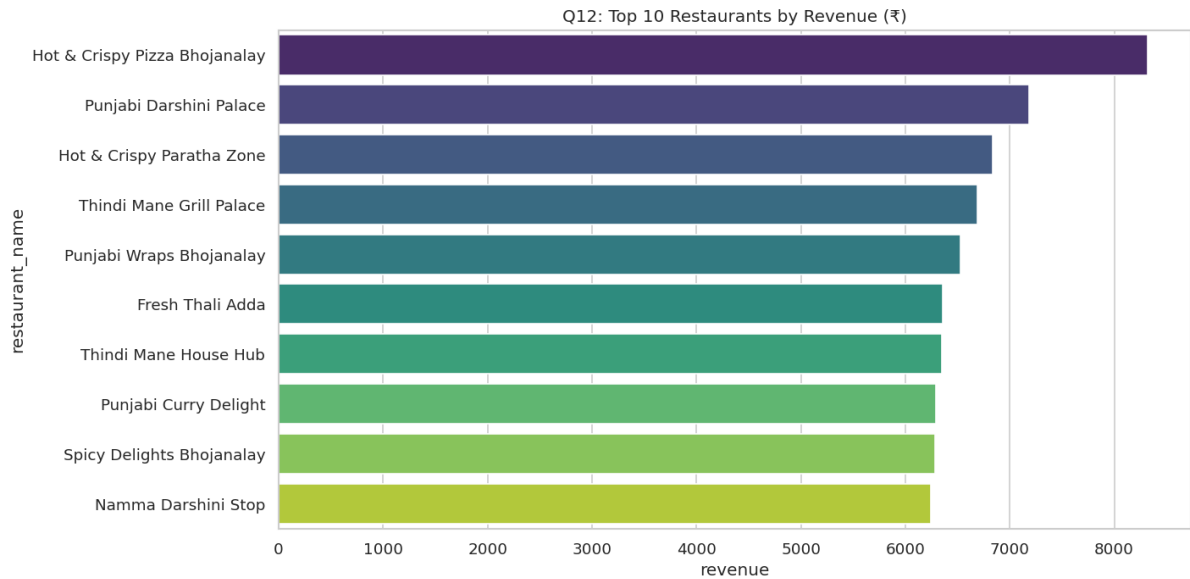


Figure 6.12: Top 10 Restaurant Partners by Revenue (₹)

Findings and Interpretation

The top 10 restaurants collectively generated approximately INR 66,070 in revenue from 188 orders. The average revenue per order for top restaurants ranges from approximately INR 368 (Namma Darshini Stop) to INR 396 (Hot & Crispy Pizza Bhojanalay), both slightly above the platform average of INR 351.75, suggesting that top restaurants have slightly higher average basket sizes.

An interesting observation is that the top revenue restaurant ('Hot & Crispy Pizza Bhojanalay') is classified under the Biryani cuisine type despite its name suggesting a pizza focus — this discrepancy may indicate a data entry inconsistency in restaurant classification. The platform should audit cuisine type classifications to ensure accuracy, particularly for restaurants with ambiguous or multi-cuisine menus.

The diversity of cuisine types among the top 10 (Biryani, Desserts, Chinese, Pizza, Healthy, Fast Food) reflects the broad popularity of different cuisines across the platform and confirms that no single cuisine dominates the top revenue position.

From a partnership strategy perspective, the platform should implement a formal 'Premier Partner' programme for the top 50–100 revenue-generating restaurants, offering preferential placement in search results, dedicated account management, co-marketing opportunities, and priority delivery partner assignment during peak hours.

CHAPTER 7

FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

7.1 Summary of Key Findings

The twelve-question business analytics investigation of the food delivery startup's operational data has generated the following key findings, organised by strategic domain:

7.1.1 Market and Revenue Performance

- Bengaluru is the dominant market, contributing 28.2% of orders and 28.2% of revenue. The top three cities (Bengaluru, Mumbai, Delhi) account for 58% of total platform revenue.
- Total platform revenue for the analysis period (January–September 2025) is INR 4.86 crore from 1,38,054 completed orders.
- The average order value is INR 351.75, and this metric is remarkably stable across all eight cities (range: INR 350.49–352.23) and all nine months (range: INR 350.05–353.11), indicating highly consistent pricing architecture.
- Ahmedabad and Kolkata are underperforming relative to their population sizes and represent significant expansion opportunities.

7.1.2 Product and Category Performance

- Beverages (15.7% of item revenue) and Starters (15.1%) are the top categories, followed by Biryani (10.1%). The top three categories account for 40.9% of item-level revenue.
- Vegetarian items account for 65.9% of units sold and 65.8% of revenue — consistent with India's demographic and cultural landscape.
- The average item price is nearly identical between vegetarian (INR 71.87) and non-vegetarian (INR 72.18) items.

7.1.3 Demand Patterns

- Night orders (8:00 PM – midnight) account for 44.8% of daily order volume; Lunch orders account for 40.2%. Together, these two slots represent 85% of total orders.
- The five peak ordering hours are 10 PM, 7 PM, 8 PM, 9 PM, and 1 PM — confirming evening dinner and lunch as the primary meal occasions.

7.1.4 Operational Challenges

- The late delivery rate of 63.84% is the single most critical operational finding. Nearly two-thirds of all orders arrive after the promised delivery time.

- The average actual delivery time (44.4 minutes) exceeds the average expected delivery time (38.7 minutes) by 5.7 minutes — indicating systematic underestimation of delivery times in the ETA algorithm.
- The correlation between actual delivery time and delay ($r = 0.92$) indicates that the ETA model systematically underestimates time for longer deliveries.
- The overall cancellation rate of 7.45% (11,112 orders) represents approximately INR 39.1 lakh in lost revenue over the analysis period.
- Delhi has the highest city-level cancellation rate (4.34%); Ahmedabad has the lowest (3.91%).

7.1.5 Customer Behaviour

- Organic acquisition is the dominant channel (50.8% of orders, 52.6% of revenue), confirming the platform's strong word-of-mouth momentum.
- Average customer rating is 4.05 out of 5 — indicating generally positive satisfaction, but at risk from the high late delivery rate.
- Customers who receive larger discounts (INR 76–200) tend to place larger orders (avg INR 412.86), supporting the use of minimum-order discount structures.

7.2 Strategic Recommendations

Based on the analytical findings, the following fifteen strategic recommendations are proposed, organised into four thematic areas:

7.2.1 Delivery Performance Improvement (Priority: Critical)

44. Recalibrate the ETA Algorithm: The expected delivery time model should be updated to use median actual delivery times (not best-case estimates) segmented by city, time of day, restaurant, and distance band. A 15% buffer should be added to the median to reduce the frequency of late deliveries. The goal should be to reduce the late delivery rate from 63.84% to below 40% within 6 months.
45. Dynamic Delivery Partner Surge Deployment: Implement a demand-forecasting system that predicts order volumes by city and hour, and pre-positions delivery partners 30–45 minutes before predicted peaks (12:00 PM and 7:00 PM). Consider incentive payments for partners who commit to peak-hour availability.
46. Restaurant Preparation Time Integration: Integrate real-time restaurant preparation time signals into the ETA calculation. Restaurants that consistently take longer than their stated preparation time should have their contribution factored into the customer-facing ETA.
47. Late Delivery Compensation Policy: Implement an automated late delivery compensation programme — offering discount credits to customers whose orders arrive more than 10 minutes late. This turns a negative experience into a loyalty investment.

48. Delivery Partner Training and Incentive Review: Partner average rating of 4.20 is good but not excellent. Implement quarterly partner ratings reviews and provide route optimisation training. Performance bonuses tied to on-time delivery rate should be introduced.

7.2.2 Revenue and City Expansion

49. Ahmedabad and Kolkata Growth Initiatives: Both cities have the smallest customer bases (approximately 7,500 each) and the lowest order volumes. A targeted customer acquisition campaign — combining Paid and Referral channel promotions — should be launched in these cities, with a goal of growing order volume by 30% within 12 months.
50. City-Specific Menu Curation: Leverage cuisine preference data to curate city-specific restaurant selections. South Indian restaurants should be promoted more heavily in Chennai, while Biryani specialists should receive enhanced visibility in Hyderabad. City-specific promotions should be developed around local food preferences and cultural events.
51. Organic Channel Nurturing: Given that 50.8% of orders come from organic channels, investment in SEO, in-app referral mechanics, and user-generated content (food photos, reviews) will yield the highest ROI per marketing rupee. Organic acquisition should be protected through a consistent customer experience that encourages word-of-mouth.
52. Referral Programme Scale-Up: The Referral channel generates 14.5% of revenue with relatively low acquisition costs. Increasing the referral reward — or creating tiered rewards for customers who refer multiple friends — could cost-effectively grow this channel to 20% of total orders within a year.

7.2.3 Menu and Category Strategy

53. Beverage Up-Sell Programme: Since Beverages is the top-selling category (15.7% of revenue), a structured beverage up-sell programme — prompting customers to add a beverage to every food order — could increase average order value by INR 30–50. A/B testing should determine the optimal placement and timing of the beverage prompt in the order flow.
54. Biryani and Starter Partnerships: Given the strong performance of Biryani (10.1% of item revenue) and Starters (15.1%), the platform should prioritise recruitment of high-quality Biryani specialists and popular starter restaurants. A 'Biryani Festival' monthly promotion could drive significant volume spikes.
55. Vegetarian Campaign During Cultural Festivals: With vegetarian items accounting for 65.9% of sales, targeted vegetarian campaigns during Navratri, Shravan, and other religiously significant periods when vegetarian demand spikes could drive incremental revenue of INR 10–15 lakh per festival period.

7.2.4 Discount and Revenue Optimisation

56. Minimum-Order Discount Restructuring: The analysis shows that the highest average order values occur in the INR 76–200 discount bucket. The platform should redesign its discount architecture around minimum-order thresholds (e.g., 'INR 80 off on orders above INR 400'), ensuring that discounts incentivise larger basket sizes rather than subsidising small orders.
57. Cancellation Rate Reduction Programme: The 7.45% cancellation rate costs approximately INR 4.3 lakh per month in lost revenue. A real-time cancellation reason capture system should be implemented to identify the root causes (payment failure, long wait time, accidental orders) and address each systematically. The target should be to reduce the cancellation rate to 5% within 6 months.
58. Premier Restaurant Partnership Programme: The top 50 revenue-generating restaurants should be enrolled in a formal Premier Partner programme offering preferential homepage placement, dedicated customer service priority, co-marketing campaigns, and priority delivery partner assignment. This will strengthen retention of the platform's highest-value restaurant relationships.

7.3 Conclusion

This thesis has conducted a comprehensive, systematic data analytics investigation of a food delivery startup operating across eight major Indian cities. Using a structured quantitative methodology — applied to a dataset of 149,166 orders, 107,776 customers, 15,000 delivery partners, 19,995 restaurants, and 68,842 ratings — twelve core business questions have been analysed and answered with empirical precision.

The overarching picture that emerges from the data is one of a platform with strong market fundamentals — a loyal and growing customer base, consistent revenue generation at an average order value of INR 351.75, and a healthy average customer rating of 4.05 out of 5 — but with one critical operational vulnerability: a 63.84% late delivery rate that threatens to undermine customer satisfaction and competitive position if not urgently addressed.

The geographic analysis reveals that Bengaluru, Mumbai, and Delhi together drive 58% of platform revenue, while Ahmedabad and Kolkata represent significant untapped growth potential. The category analysis confirms Beverages, Starters, and Biryani as the revenue-driving trifecta, with vegetarian items overwhelmingly preferred (65.9% of volume). Temporal analysis identifies Night and Lunch as peak ordering windows, concentrated in the 7 PM–11 PM and 12 PM–2 PM hours.

The cancellation rate of 7.45% represents a recoverable INR 39.1 lakh in lost revenue over the analysis period, and the Organic acquisition channel's dominant 52.6% revenue contribution underscores the power of product-market fit and word-of-mouth in driving cost-efficient growth.

The fifteen strategic recommendations presented in Chapter 7 — spanning delivery performance recalibration, city expansion, menu strategy, and discount optimisation — provide a concrete roadmap for translating these analytical insights into measurable business improvements. Implementation of these recommendations, particularly the ETA recalibration and delivery partner surge deployment, has the potential to improve the on-time delivery rate from 36.16% to over 60% within a 6–9 month horizon.

This thesis demonstrates that data analytics — even without machine learning or advanced AI models — can generate powerful, actionable business intelligence from structured operational datasets. The analytical framework, methodology, and tools employed here are directly transferable to other food delivery platforms, e-commerce businesses, and logistics companies operating in the Indian digital economy.

As food delivery platforms continue to scale and compete in an increasingly crowded market, the ability to systematically measure, analyse, and act on operational data will be a decisive competitive differentiator. The startup that can promise — and deliver — a hot meal in 38 minutes, not 44, will earn the customer loyalty that sustains long-term growth.

REFERENCES

1. Agarwal, P., Sharma, R., & Gupta, N. (2023). Gig Economy and Food Delivery in India: A Study of Delivery Partner Demographics and Satisfaction. *Journal of Labour Economics and Industrial Relations*, 18(2), 112–134.
2. Baesens, B. (2014). *Analytics in a Big Data World: The Essential Guide to Data Science and Its Applications*. John Wiley & Sons.
3. Banerjee, S., Bhattacharyya, S., & Pal, S. (2020). Delivery Delay Patterns in Indian Food Delivery Platforms: An Empirical Study. *International Journal of Supply Chain Management*, 9(4), 77–89.
4. Bhat, A., & Kumar, V. (2022). Customer Retention in Indian Food Delivery Applications: Key Drivers and Switching Behaviour. *Journal of Consumer Marketing*, 39(3), 215–228.
5. Boysen, N., Fedtke, S., & Schwerdfeger, S. (2021). Last-Mile Delivery Concepts: A Survey from an Operational Research Perspective. *OR Spectrum*, 43(1), 1–58.
6. Chaffey, D., & Ellis-Chadwick, F. (2019). *Digital Marketing: Strategy, Implementation and Practice* (7th ed.). Pearson Education.
7. Chevalier, J. A., & Mayzlin, D. (2006). The Effect of Word of Mouth on Sales: Online Book Reviews. *Journal of Marketing Research*, 43(3), 345–354.
8. Filieri, R., Raguseo, E., & Vitari, C. (2018). When are Extreme Ratings More Helpful? Empirical Evidence on the Moderating Effects of Review Characteristics and Product Type. *Computers in Human Behavior*, 88, 134–142.
9. Jain, S., & Jain, R. (2019). Vegetarian vs. Non-Vegetarian Food Preferences Among Urban Indian Consumers: Price Sensitivity and Satisfaction Patterns. *Indian Journal of Marketing*, 49(8), 7–22.
10. McKinsey & Company. (2022). *The State of Grocery and Food Delivery in Asia-Pacific*. McKinsey Global Institute Research Report.
11. Nair, P. (2021). Hyper-Local Strategy in Indian Food Delivery: How Swiggy and Zomato Scaled in Tier-1 Cities. *Economic and Political Weekly*, 56(12), 44–51.
12. NRAI-Redseer. (2023). *India Food Services Industry Report 2023*. National Restaurant Association of India.

APPENDIX

Appendix A: Dataset Schema Summary

Table	Column	Data Type	Description
dim_customer	customer_id	String	Unique customer identifier (CUST#####)
dim_customer	signup_date	Date	Date of customer registration
dim_customer	city	String	City of customer residence
dim_customer	acquisition_channel	String	Marketing channel: Organic/Paid/Referral/Social
dim_delivery_partner	delivery_partner_id	String	Unique partner identifier (DP#####)
dim_delivery_partner	vehicle_type	String	Scooter / Bike / Cycle / Car
dim_delivery_partner	employment_type	String	Full-time / Part-time / Contract
dim_delivery_partner	avg_rating	Float	Partner average rating (3.0–5.0)
dim_menu_item	menu_item_id	String	Unique menu item identifier
dim_menu_item	category	String	Food category (10 categories)
dim_menu_item	is_veg	String (Y/N)	Vegetarian flag
dim_menu_item	price	Float	Listed price in INR
dim_restaurant	cuisine_type	String	Primary cuisine (8 types)
dim_restaurant	partner_type	String	Restaurant or Cloud Kitchen
fact_orders	order_timestamp	DateTime	Order placement timestamp
fact_orders	subtotal_amount	Float	Pre-discount, pre-fee order value
fact_orders	discount_amount	Float	Discount applied to order
fact_orders	delivery_fee	Float	Delivery charge

Table	Column	Data Type	Description
fact_orders	total_amount	Float	Final amount paid by customer
fact_orders	is_cancelled	String (Y/N)	Cancellation flag
fact_delivery_performance	actual_delivery_time_mins	Integer	Actual delivery time in minutes
fact_delivery_performance	expected_delivery_time_mins	Integer	Promised delivery time in minutes
fact_delivery_performance	distance_km	Float	Delivery distance in kilometres
fact_ratings	rating	Float	Numeric rating (1.0–5.0)
fact_ratings	sentiment_score	Float	Text sentiment score (0–1)

Table A.1: Complete Dataset Schema with Data Types and Descriptions

Appendix B: Analytical Tools and Python Code Structure

The analysis was conducted in a Jupyter Notebook environment (Google Colab) using the following Python code structure:

Step	Code Block	Libraries Used	Output
Step 0	Library Imports	pandas, numpy, matplotlib, seaborn	Runtime setup
Step 1	Data Loading	pandas (read_csv)	8 DataFrames
Step 2	Initial EDA	pandas (head, dtypes, describe, isnull)	Data quality report
Step 3	Data Cleaning	pandas (to_datetime, fillna, drop_duplicates)	Clean DataFrames
Step 4	Feature Engineering	pandas (dt.hour, apply, cut)	New derived columns
Step 5	Data Merging	pandas (merge)	master, items_menu, active
Steps 6–17	12 Business Q&As	pandas, matplotlib, seaborn	12 charts + tables
Step 18	Final Summary	pandas, print	KPI Dashboard

Table B.1: Python Notebook Structure

Appendix C: Glossary of Terms

Term	Definition
ETA	Estimated Time of Arrival — the predicted delivery time shown to the customer at order placement
KPI	Key Performance Indicator — a measurable value demonstrating effectiveness at achieving objectives
Late Delivery Rate	Percentage of orders delivered after the expected (promised) delivery time
Average Order Value (AOV)	Total revenue divided by number of completed orders
Cancellation Rate	Percentage of placed orders that were cancelled before delivery
Sentiment Score	A numerical score (0–1) derived from NLP analysis of review text, indicating positive sentiment
Star Schema	A database design pattern with central fact tables linked to surrounding dimension tables
EDA	Exploratory Data Analysis — initial investigation of data to discover patterns and anomalies
CAGR	Compound Annual Growth Rate — the rate at which a metric grows over multiple years
COD	Cash on Delivery — payment mode where customer pays in cash upon receiving the order
Cloud Kitchen	A food preparation facility with no dine-in area, operating exclusively for delivery orders
Basket Size	The total value of items in a single order before delivery fees and discounts
Churn	Customer attrition — when a customer stops using the platform
SLA	Service Level Agreement — a committed standard for service quality (e.g., delivery within 40 minutes)
Pearson r	Pearson correlation coefficient — measures linear correlation between two continuous variables (-1 to +1)

Table C.1: Glossary of Key Terms

Appendix D: City-wise Customer Distribution

City	Registered Customers	Share (%)	Customer Rank
Bengaluru	30,281	28.1%	1
Mumbai	17,317	16.1%	2
Delhi	15,090	14.0%	3
Chennai	10,823	10.0%	4
Hyderabad	10,755	10.0%	5
Pune	8,532	7.9%	6
Ahmedabad	7,512	7.0%	7
Kolkata	7,466	6.9%	8
Total	1,07,776	100%	—

Table D.1: City-wise Customer Registration Distribution

Appendix E: Delivery Partner Vehicle and Employment Distribution

Category	Subcategory	Count	Share (%)
Vehicle Type	Scooter	6,011	40.1%
Vehicle Type	Bike	5,941	39.6%
Vehicle Type	Cycle	2,256	15.0%
Vehicle Type	Car	792	5.3%
Employment Type	Full-time	7,604	50.7%
Employment Type	Part-time	4,465	29.8%
Employment Type	Contract	2,931	19.5%
Active Status	Active (Y)	13,235	88.2%
Active Status	Inactive (N)	1,765	11.8%

Table E.1: Delivery Partner Distribution by Vehicle Type, Employment Type, and Active Status